Sonogashira Coupling

Dr. Zhang Sheng

Part 1

Theoretical Background

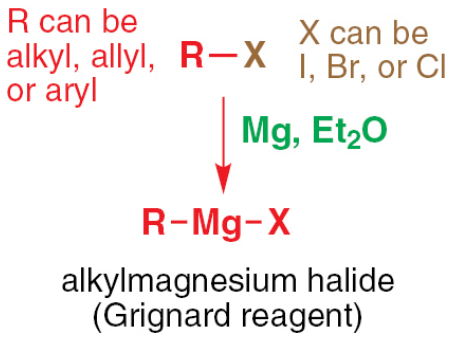
Organometallic

- Organometallic compounds: having carbon-metal/metalloid bond



- Several Nobel Prizes are awarded for organometallic chemistry

Nobel Prizes – Organometallic





Victor Grignard
(1912)
Grignard reagent

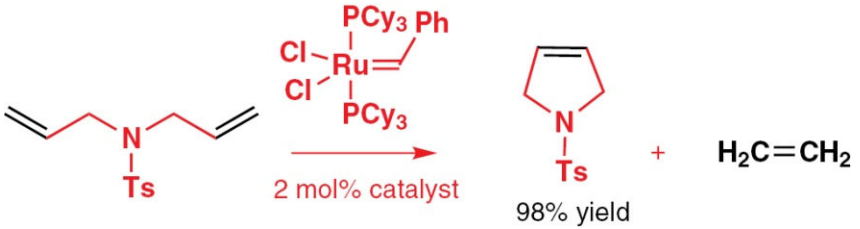


ferrocene

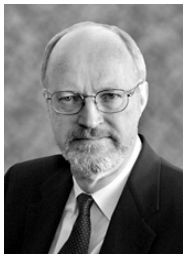




Ernst Otto Fischer Geoffrey Wilkinson
(1973)
Sandwich compounds









Yves Chauvin Robert H. Grubbs Richard R. Schrock
(2005)
Alkene metathesis

Nobel Prize – Organometallic (2010)

- Jointly to Richard F. Heck, Ei-ichi Negish and Akira Suzuki “for palladium-catalyzed cross couplings in organic synthesis”.



Richard F. Heck



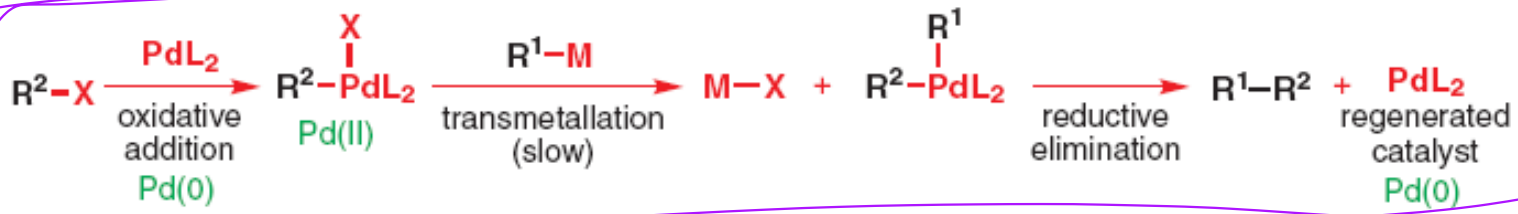
Ei-ichi Negishi



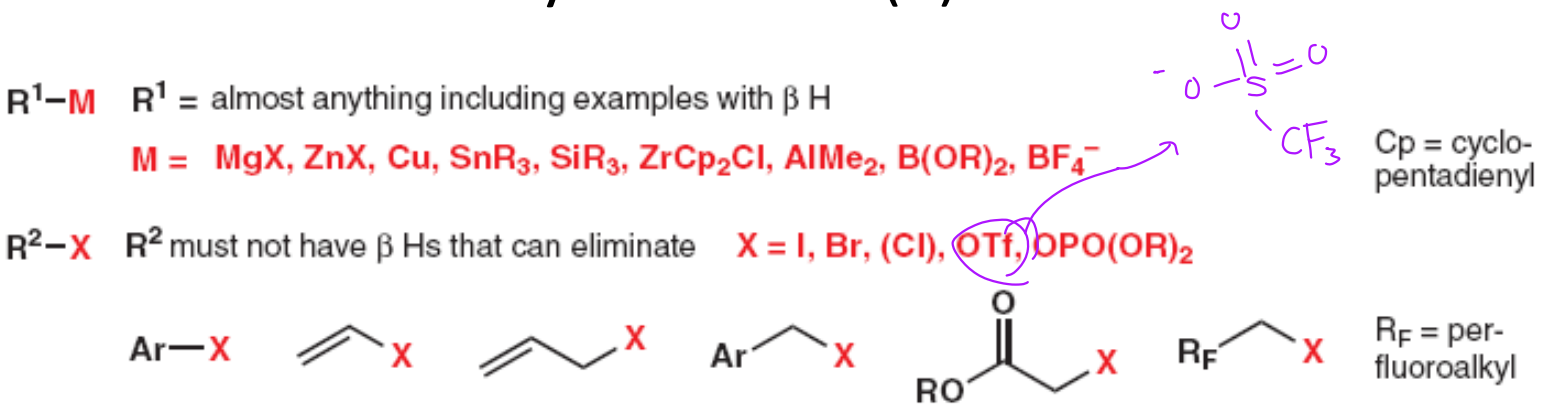
Akira Suzuki

Pd-catalyzed Cross-coupling

- General mechanism:

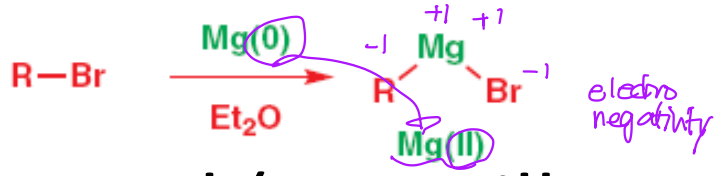


  - Oxidative addition: Pd(0) → Pd(II)
  - Transmetallation: exchange of R^1 and X
  - Reductive elimination: Pd(II) → Pd(0)
  - Restart another cycle with Pd(0)

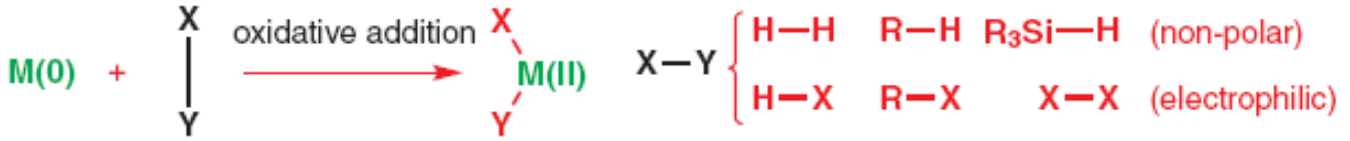


Oxidative Addition

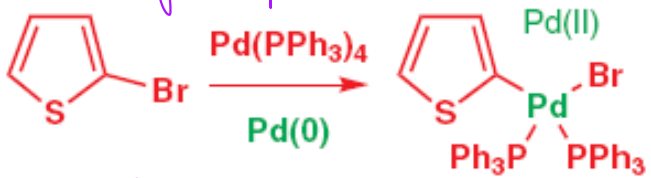
- Oxidation state of metal is increased by two
  - For example: M(0) → M(II)



- Coordinatively unsaturated metal (can still accept ligand) → Coordinatively saturated metal (cannot accept more ligand)



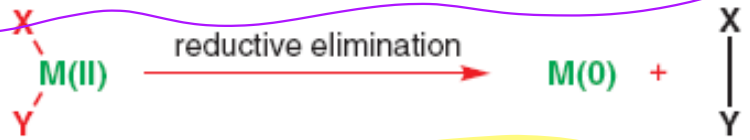
'ligand displacement'. 2 PPh_3 leaves & then bond 로 C & Br



how to do e⁻ count?
(inorganic 파트 참고)

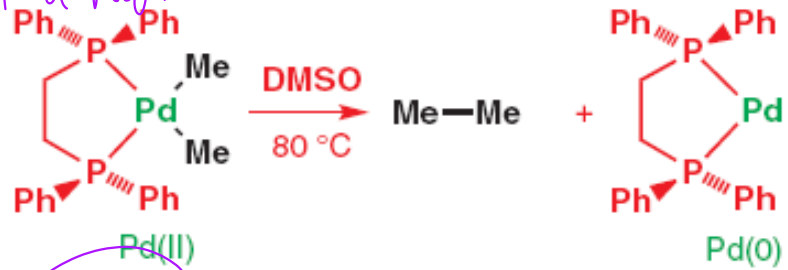
Reductive Elimination

- Reverse of oxidative addition
- Oxidation state of metal decreases by two
  - For example: M(II) → M(0)
- A new σ bond is formed between the two ex-ligands

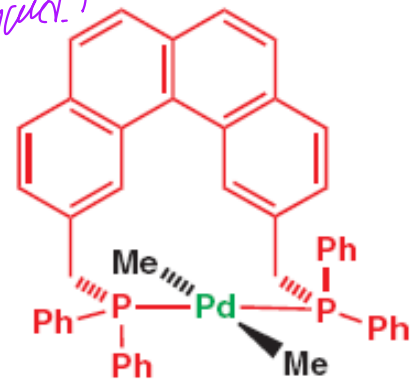


- These two ligands must be *cis* to be eliminated

* for reductive E' to occur, Ligands must be 'cis' to be present!
↳ not a requirement for oxidative addition.



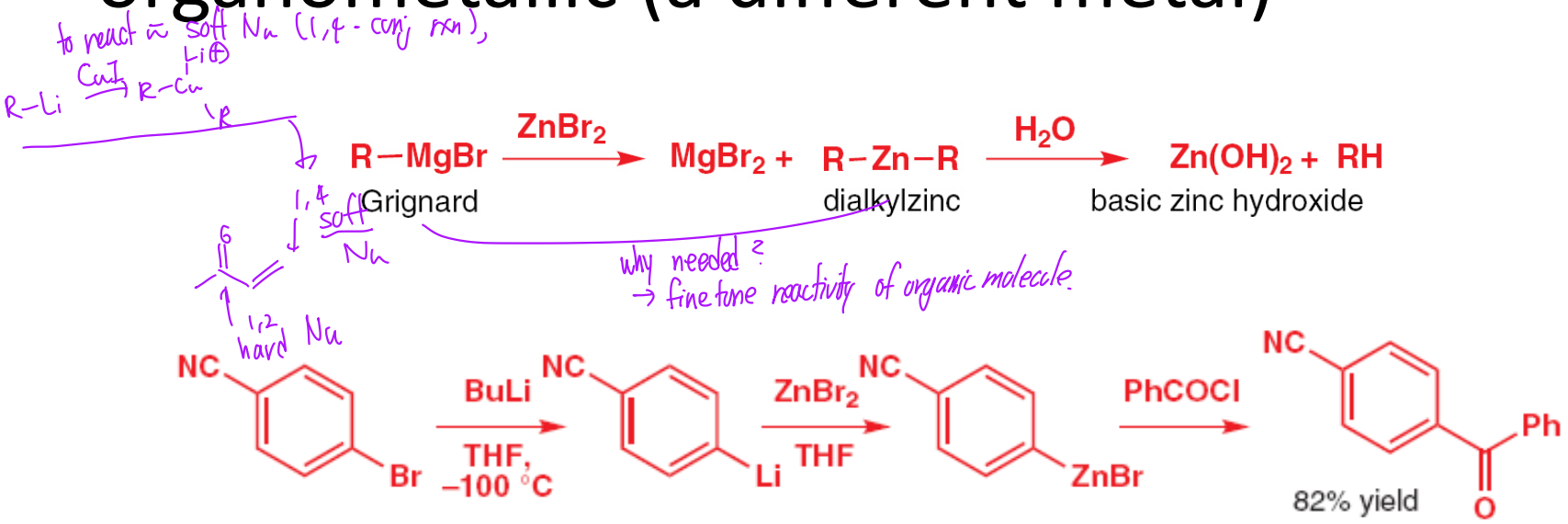
ligands *cis*: reductive elimination

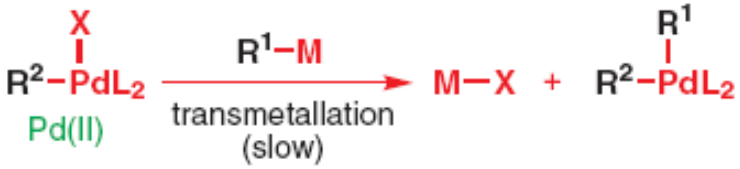


ligands *trans:* no reaction

Transmetallation

- One type of organometallic → another type of organometallic (a different metal)



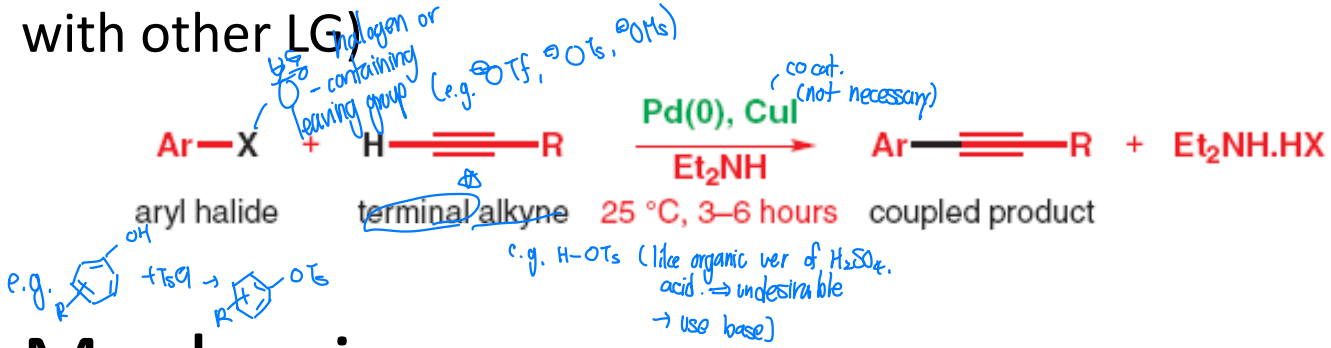


Sonogashira Coupling

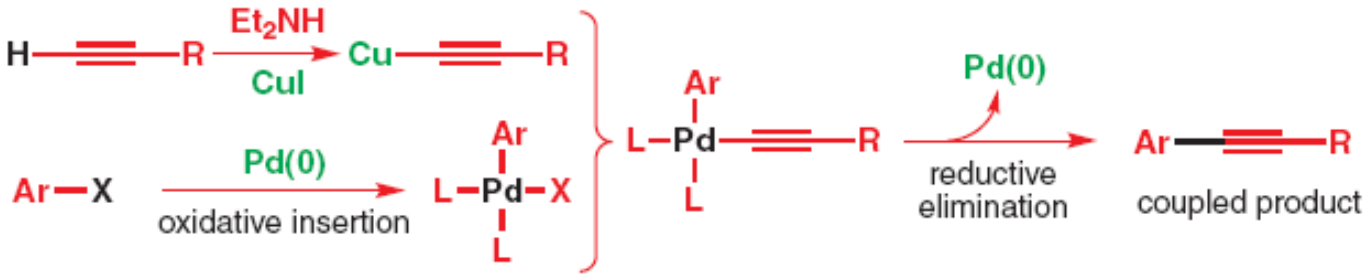


Kenkichi Sonogashira

- Terminal alkyne + aryl/vinyl halide (or with other LG)



- Mechanism:
  - Oxidative addition → Transmetallation → Reductive elimination

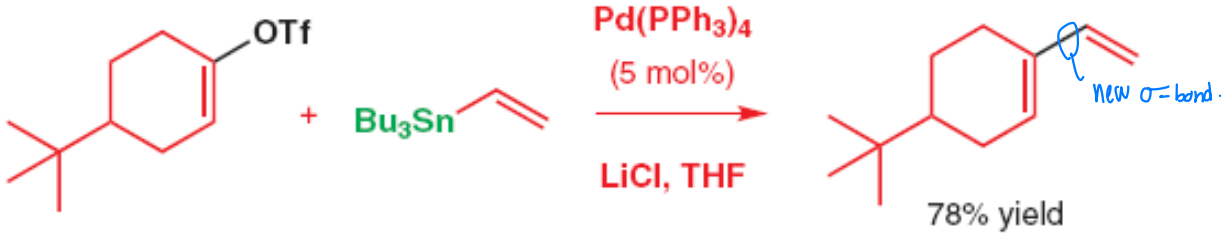


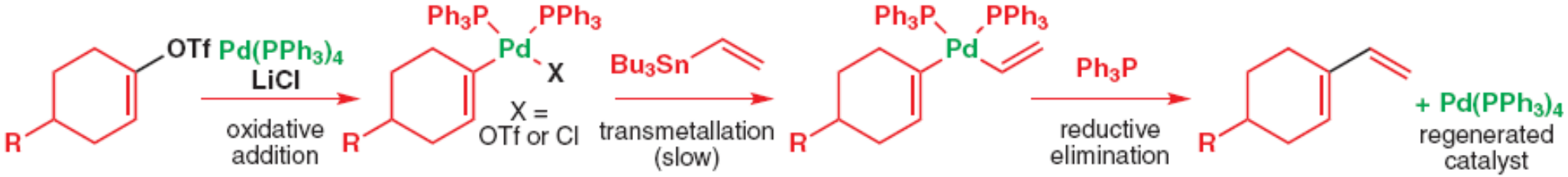
Stille Coupling

- Organotin (R^1Sn)+ Aryl/vinyl halide



John Kenneth Stille (died in 1989, before the Nobel Prize)



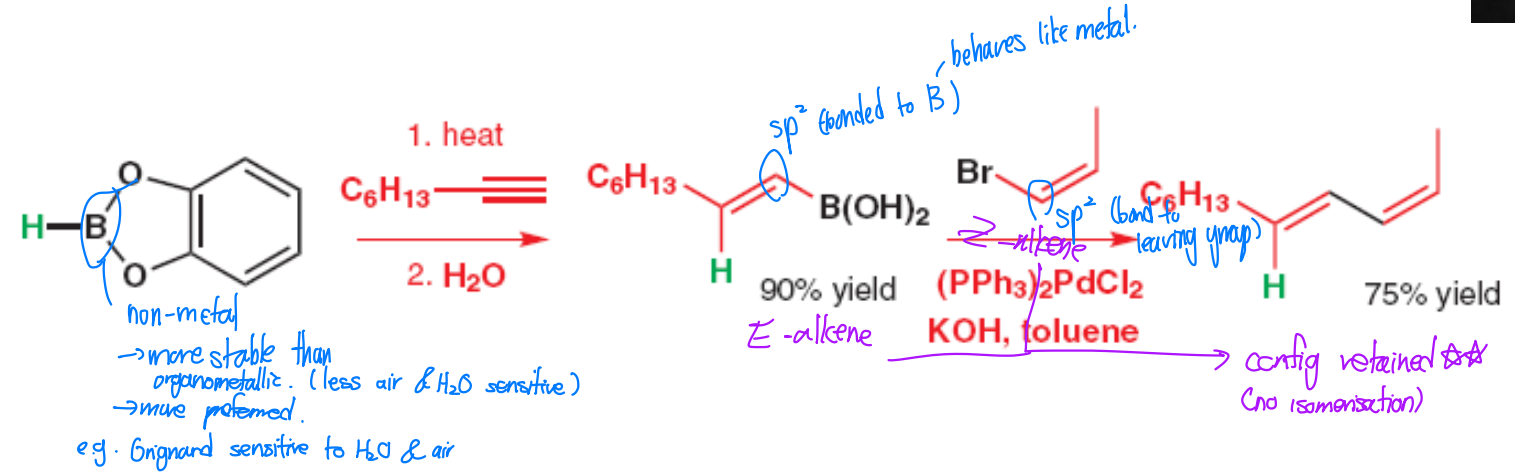


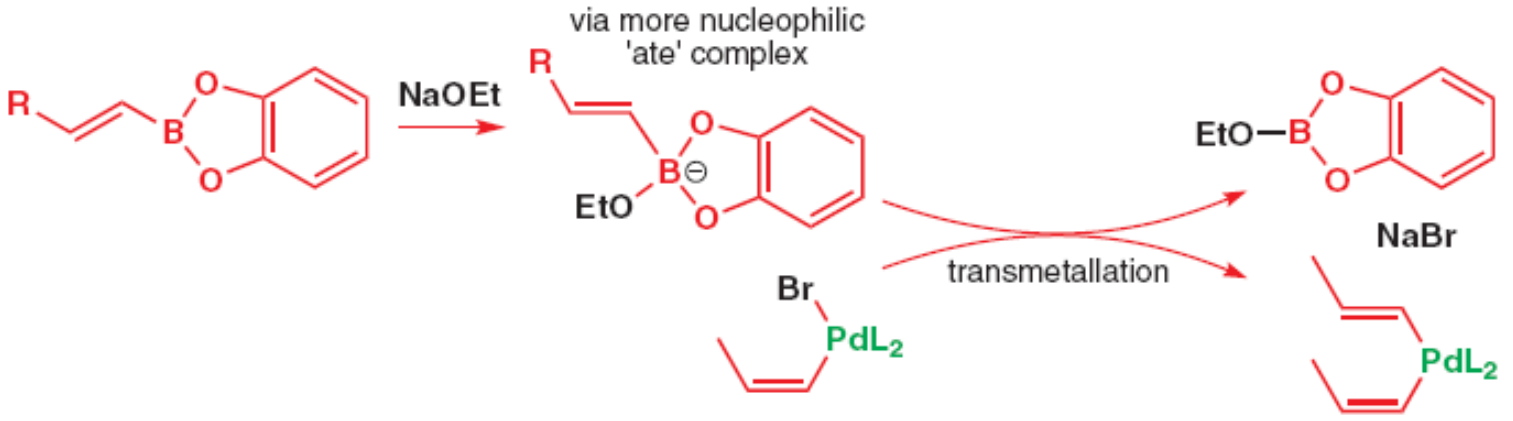
Suzuki Coupling



Akira Suzuki

- Boronic acid + aryl/vinyl halide



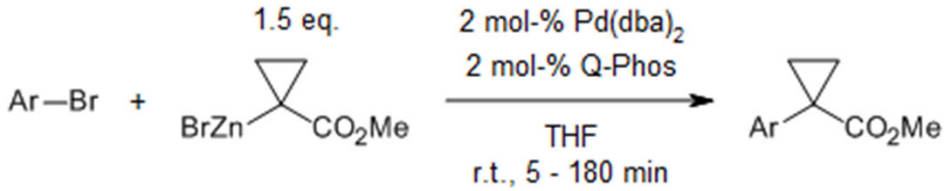


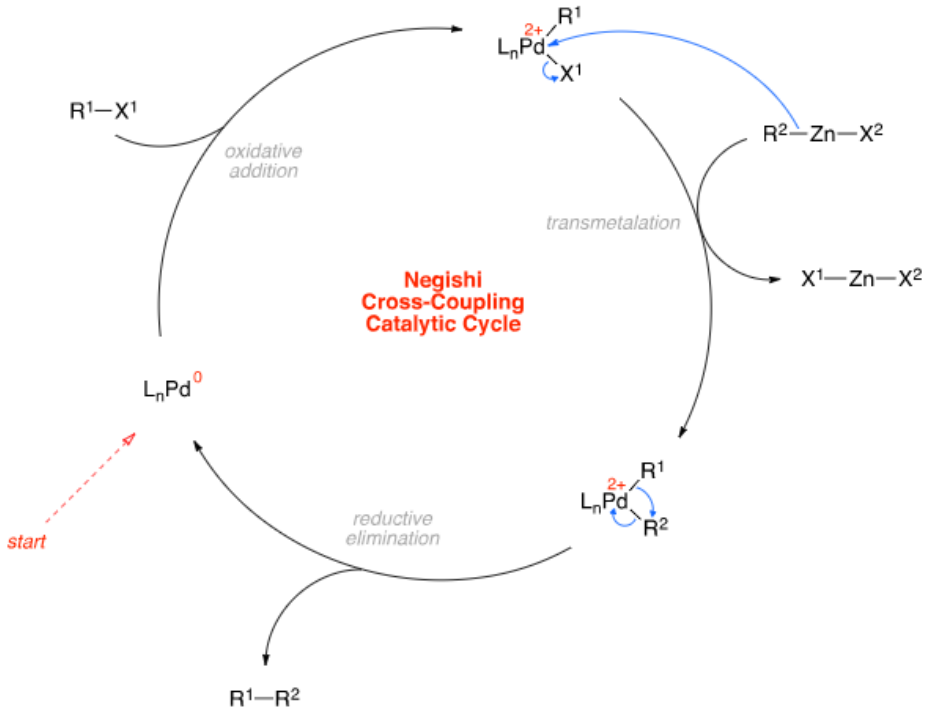
Negishi Coupling

- Organozinc (R^1ZnX) + aryl/vinyl halide



Ei-ichi Negishi





*sonogashira : sp^2 & sp rxn
*for sp^3, conventional S_N1/S_N2 is suff for forming new σ-bond

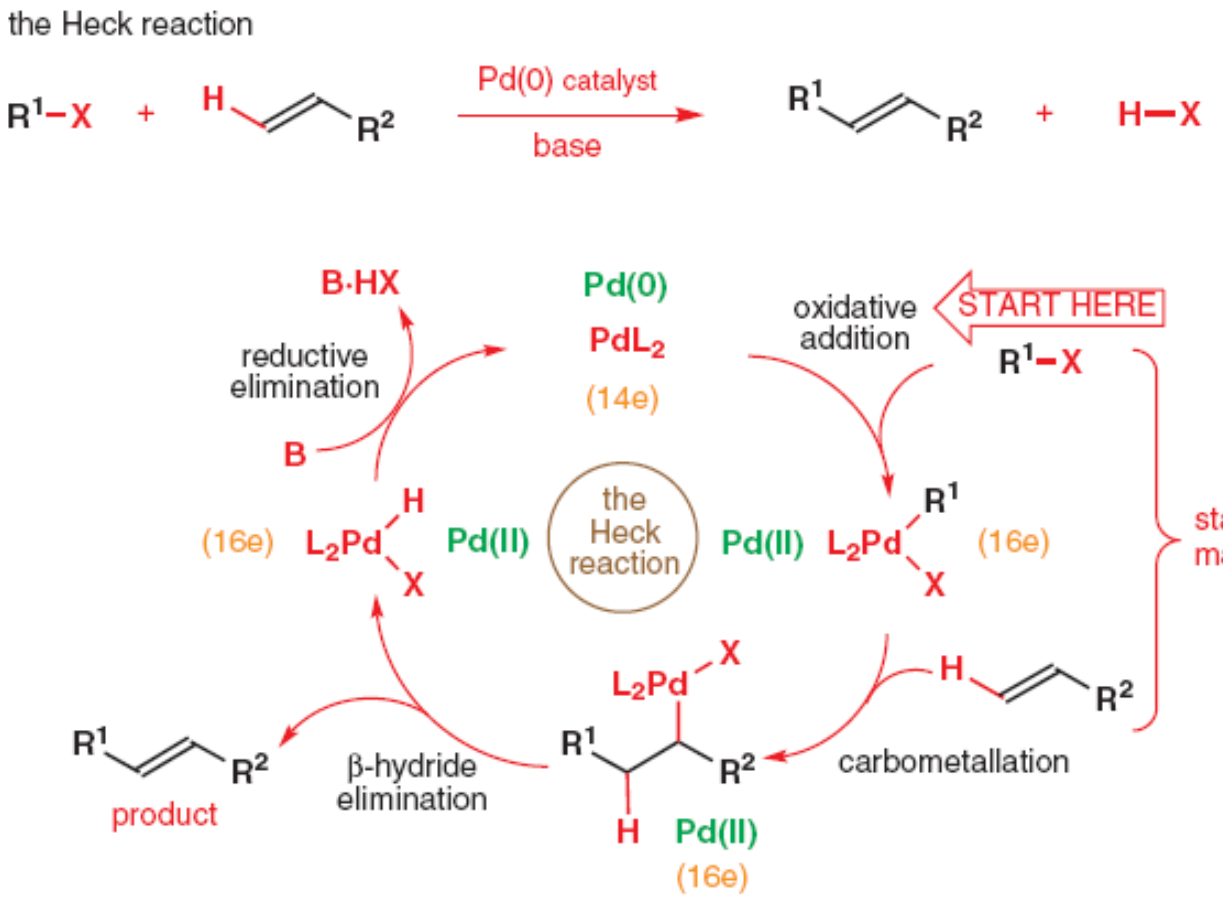
Heck Reaction

└ no organometallic compound used. just Pd(0) cat.

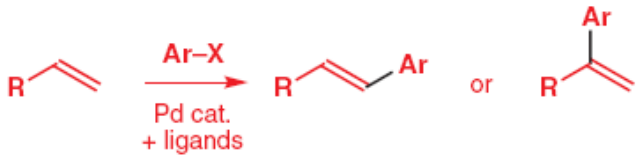
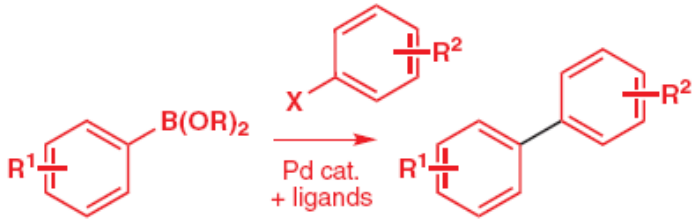
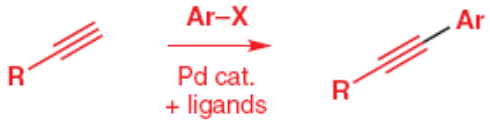
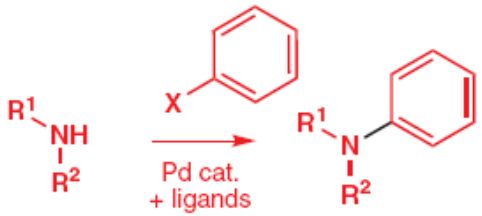
- Alkene + aryl/vinyl halide



Richard F. Heck



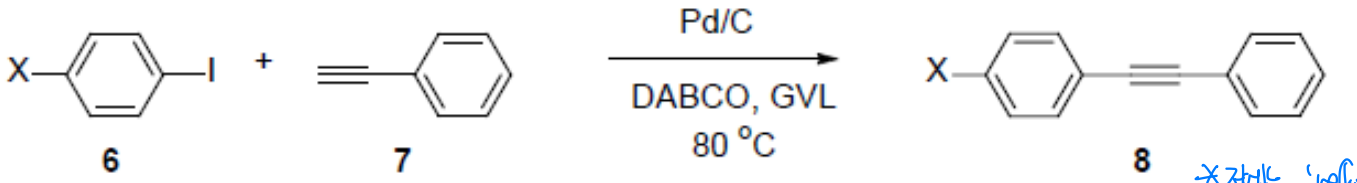
Summary of most common coupling

Coupling an aryl or vinyl halide with...	Typical example (X=I, Br, OTf)	See page	Name of reaction
an alkene		1079	Heck
an aryl or vinyl stannane		1084	Stille
an aryl or vinylboronic acid or ester		1085	Suzuki
an alkyne		1087	Sonogashira
an amine		1092 (later in this chapter)	Buchwald-Hartwig

Part 2

Practical

Reaction Setup (1)



6 7 8

X = NO_2, H, COMe, COOMe ...

- Turn on the heater equipped with metal heating block and set the temperature to 80 °C. The heater will be shared since one set allows multiple reactions to be conducted simultaneously with accurate temperature control.
- Collect an unknown aryl iodide (1.0 mmol, pre-weighed in glass vial) and the Pd/C catalyst (10 wt%, 5.3 mg, in small plastic Eppendorf tube) from the laboratory staff.
- Withdraw trace amount of the aryl iodide for TLC analysis later.

REMEMBER!!!

Reaction Setup (2)

- Transfer all the Pd/C catalyst into the glass vial (this is your reaction vessel). To the glass vial, add DABCO (0.168 g, 1.5 mmol, 1.5 equiv.) and a small magnetic stirrer.
- Measure 1.0 mL of GVL solvent and use it to rinse the vial, make sure all solid settle at the bottom of the vial. Do not shake the vial. No compound shall stick on the side wall since they will not react.
- Use a syringe to add phenylacetylene (1.5 mmol) to the reaction mixture. Close the vial and place the reaction mixture on the heating block maintained at 80 °C.

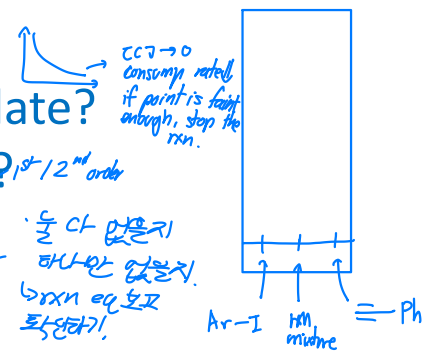
How much to measure?

Reaction Setup (3)

- Perform a TLC analysis (Appendix A) to monitor the progress of the reaction after 1h. To prepare sample for TLC analysis: dip a dropper (without rubber teat) into the mixture to withdraw a small amount of sample. Place the tip of the dropper into a clean sample vial and run about 0.5 mL of ethyl acetate down the dropper from the bulb end to dissolve the reaction mixture (why do you need to dissolve the reaction mixture in an external solvent?). Label as R1 (Reaction 1). Spot R1 against the two starting materials (aryl iodide and alkyne) using 5% ethyl acetate/95% hexane as the eluting solvent. Observe and record the TLC plate under a UV lamp. If the eluent does not give a good separation on the TLC plate, adjust the eluent ratio (2% / 1% / *ect.* ethy acetate in hexane) until a good separation is achieved. The eluent ratio which gives a good separation will be used in Week 2 for column chromatography purification of the crude product.

dependant on the product structure

- What is your expected observation of the TLC plate?
- How to judge whether the reaction is complete?



Reaction Setup (4)



Phenylacetylene spot may not be visible on the TLC plate. Why?

└ v volatile

- If the reactant (limiting reagent) spot on the TLC plate is absent (or very faint), the reaction is complete. Otherwise, analyse the reaction mixture again by TLC after every 20-30 minutes until i) the reaction is complete, or ii) the maximum duration of the reaction (2 hours) is reached.

Maximum reaction duration: 2 hours

How to perform TLC will be taught
in the lecture "column chromatography + TLC"

Work-up (1)

Remove the reaction mixture from the heating block and cool it to room temperature. Add 10 mL of hexane: ethyl acetate (4:1) to the reaction mixture and mix well. Filter the mixture through a fluted filter paper and rinse the vial with 10 mL × 2 of the same solvent. Transfer the filtrates into a 100 mL separatory funnel.

simple gravity filtration.
× use suction

for removing insoluble Pd/C cat.

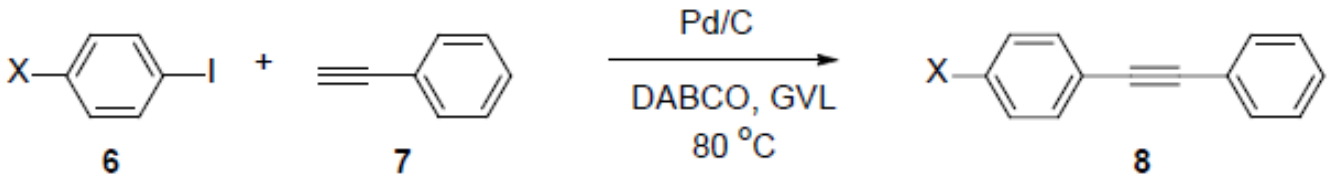
- Why filter?
  ↳ which component remain?
- Why rinse?
  ↳ which component remain?

Work-up (2)

Wash the organic layer with 10 mL of aqueous 1M HCl, 30 mL of water, and then 10 mL of brine consecutively. Dry the organic layer with anhydrous sodium sulfate. Filter the solution into a dry 100 mL round bottom flask and remove the solvent using a rotary evaporator.

Keep the crude product (inside the RBF) in designated fumehood

- What is removed in each extraction?
- How to use rotary evaporator?
- How much crude product do you expect to obtain?



6 7 8

X = NO_2, H, COMe, COOMe ...

Purification (1)

- Column chromatography
  - What is the principle?
  - What are the pros and cons?

- Packing column → Loading sample → Running column chromatography → Collection of different components and check by TLC → Rotavap to obtain the product → Clean up

Purification (2)

- Column eluent: In a 250 mL beaker, prepare 200 mL of ethyl acetate-hexane eluent. You have determined the ratio (e.g. 5% / 2% / 1% / *ect.* ethy acetate in hexane) by TLC in the previous week.
- Packing the column according to Appendix B. Be reminded that the eluent used for packing the column can be reused as column eluent for separation.

  - Wet packing
  - Packing should be dense
  - No air bubbles trapped in the packing

Purification (3)

What will happen if too much DCM is used?

- Load the sample: Dissolve the crude product in a minimum volume of dichloromethane (1-2 mL). Load the sample solution gently onto the silica gel column according to Appendix B.
- Conduct column chromatography separation following Appendix B. [Note: you should not rely on the color to decide the composition.]
- The product may have a low concentration in the eluent. You are encouraged to spot the content of each test tube multiple times on the TLC plate to ensure you do not miss the desired product.

IMPORTANT!!!
Some students disposed off their product by mistake

Purification (4)

- If the product has not been eluted out, add more eluent or increase the polarity of the eluent. Continue to run the column until the entire desired product has been eluted.

REMEMBER!!!

- Collect all fractions containing the desired product in a dry, pre-weighed round bottom flask and remove the solvent using a rotary evaporator. Bring the round bottom flask back to your hood and put the product under vacuum for another 20 minutes to remove all solvent.

Characterization

- Record the mass of the product and submit sample for ^{1}H NMR analysis.

After NMR preparation, clean and return the RBF

References

- The organometallic chemistry of transition metals, Crabtree
- Organic Chemistry, J. Clayden, N. Greeves and S. Warren
- *Chem. Rev.*, **2007**, *107* (1), pp 133–173
- *Chem. Rev.*, **2011**, *111* (3), pp 2251–2320
- *Chem. Rev.*, **2007**, *107* (3), pp 874–922
- www.nobelprize.org
- Angew. Chem. Int. Ed., **2012**, *51*, pp 5062-5085